

"Marlin EV: Front brake judder from warped rotors"

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Area: bulletins

1 Condition: Front Brake Judder on Marlin EV

Owners of the Marlin EV may report a pulsation or judder felt through the brake pedal and steering wheel during moderate-to-firm braking from highway speeds. The symptom is most pronounced once the front rotors (BR-7731) reach operating temperature after repeated stops. Technicians should verify the complaint on a road test before disassembly, noting the speed range at which the judder appears. Refer to the Marlin EV Brakes Service Manual :: Front Brakes for system layout and inspection points.

2 Cause: Warped Front Rotors

The judder is caused by excessive lateral runout and thickness variation in the front rotors (BR-7731), which develop after the original front brake pads (BP-7720) deposit friction material unevenly under sustained high-temperature braking. As the warped rotor surface passes the caliper, the resulting brake torque variation transmits the pulsation into the chassis. Confirm runout and minimum thickness against the values in the Marlin EV Brakes Service Manual :: Front Pad and Rotor Specifications before condemning the parts. Do not resurface rotors below the published minimum thickness.

3 Repair Procedure: Front Brake Pad and Rotor Service

Replace both front rotors (BR-7731) and the front brake pads (BP-7720) as a set, following PROC-BRAKE-FR to ensure correct caliper bracket torque and pad seating. Bed in the new friction material per the bedding sequence so that the BP-7720 pads transfer an even transfer layer onto the new BR-7731 rotors. The full step-by-step is documented in the Marlin EV Brakes Service Manual :: Front Brake Pad and Rotor Service Procedure . For background on the complete service flow, see the Marlin EV Brakes Service Manual .